

ADNEX2: Updating the ADNEX model to predict ovarian cancer for surgically and conservatively managed patients with an adnexal mass

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Abstract

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Objective: The ADNEX prediction model to diagnose ovarian cancer has shown good performance in many external validation studies. Our aim is to update ADNEX to ADNEX2, using recent data that also include patients that were managed with follow-up. ADNEX2 is designed for use after applying the benign descriptors (BDs), a set of ultrasound profiles with a risk of malignancy lower than 1%.

Design: Secondary analysis of an ongoing multicentre prospective cohort study.

Setting: 17 oncology and non-oncology centres from 8 European countries.

Participants: Patients with a newly detected adnexal mass recruited between 2012 and 2016 irrespective of whether they were managed surgically or with follow-up.

Interventions: Model updating involved logistic recalibration followed by Bayesian refitting. Validation was based on leave-centre-out cross-validation. Missing values were addressed with multiple imputation.

Main outcome measures: Model performance in terms of discrimination, calibration, and clinical utility to distinguish benign from malignant masses.

Results: 6847 patients were included; 3900 (57%) were immediately operated, 2639 (39%) were managed with clinical and ultrasound follow-up, and for 308 (4%) the management was unknown. The BDs applied in 2190 (32%) patients, of whom 10 (0.5%, 95% CI 0.2-0.8) had a histologically confirmed malignancy. ADNEX2 was developed on the remaining 4657 patients. On all 6847 patients, the AUC was 0.95 (95% CI 0.93-0.96) for BDs followed by ADNEX2 including CA125 if available. On 4657 patients for which BDs did not apply, ADNEX2 including CA125 if available had an AUC of 0.93 (0.91-0.94).

Conclusions: ADNEX2 is the first model in this field that is developed on patients irrespective of subsequent management. ADNEX2 targets patients for which the BDs do not apply. For the decision of

whom to refer for surgery, CA125 is not crucial. However, CA125 is strongly recommended to preoperatively assess the likely type of malignancy.

Trial registration: [clinicaltrials.gov NCT01698632](https://clinicaltrials.gov/ct2/show/study/NCT01698632).

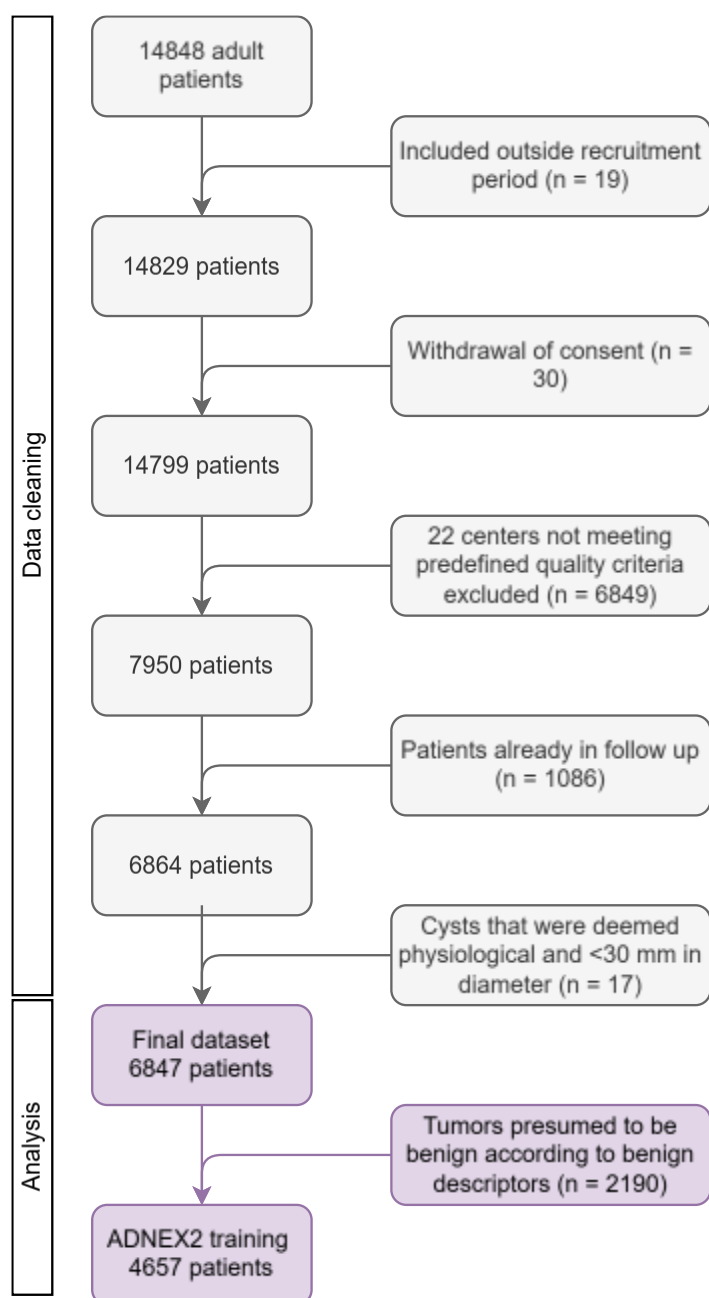


Figure 1. Patient flow diagram of IOTA5 second interim analysis

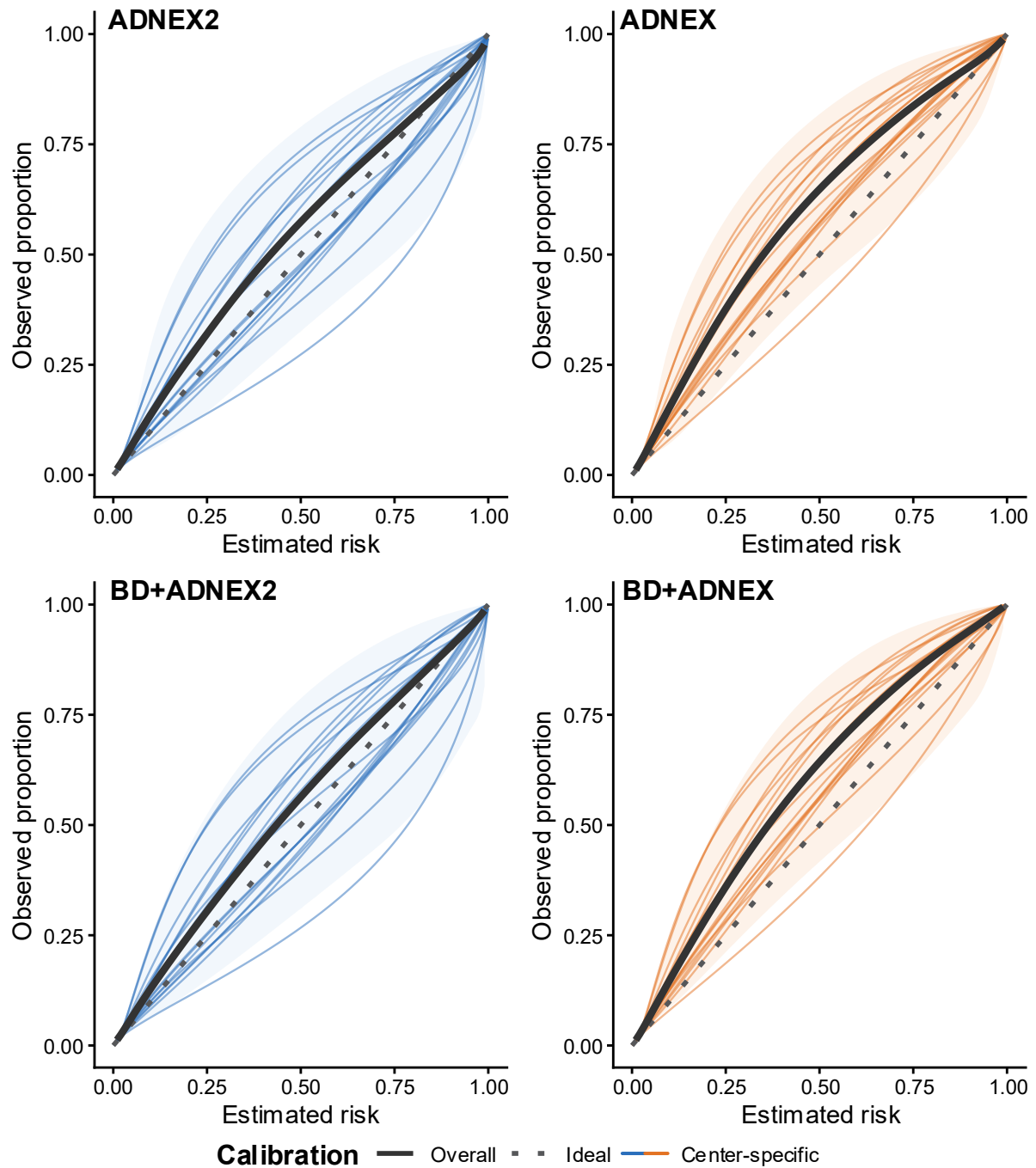


Figure 2. Clustered calibration curves for ADNEX2 without CA125 and ADNEX without CA125 if benign descriptors do not apply ($n = 4657$), and the two-step strategy with ADNEX2 without CA125 and the two-step strategy with ADNEX without CA125 in all patients ($n = 6847$). The shaded blue and orange area indicate the 95% prediction interval for the calibration.

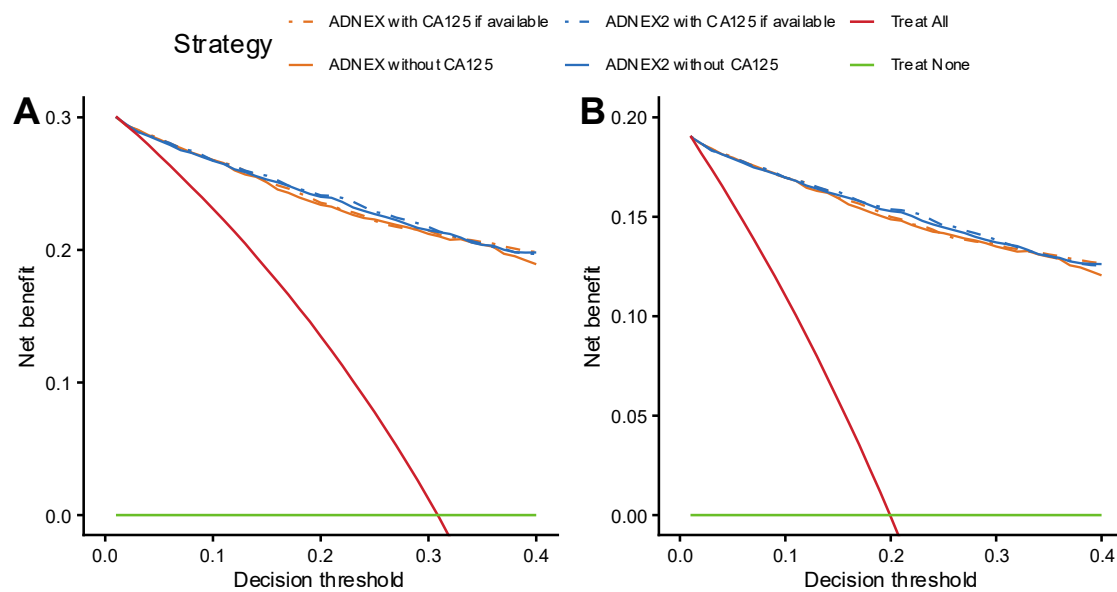


Figure 3. Decision curves for ADNEX2 and ADNEX in patients where BDs did not apply (n = 4657) (A) and in the two-step strategy with ADNEX2 or ADNEX (n = 6847) (B).

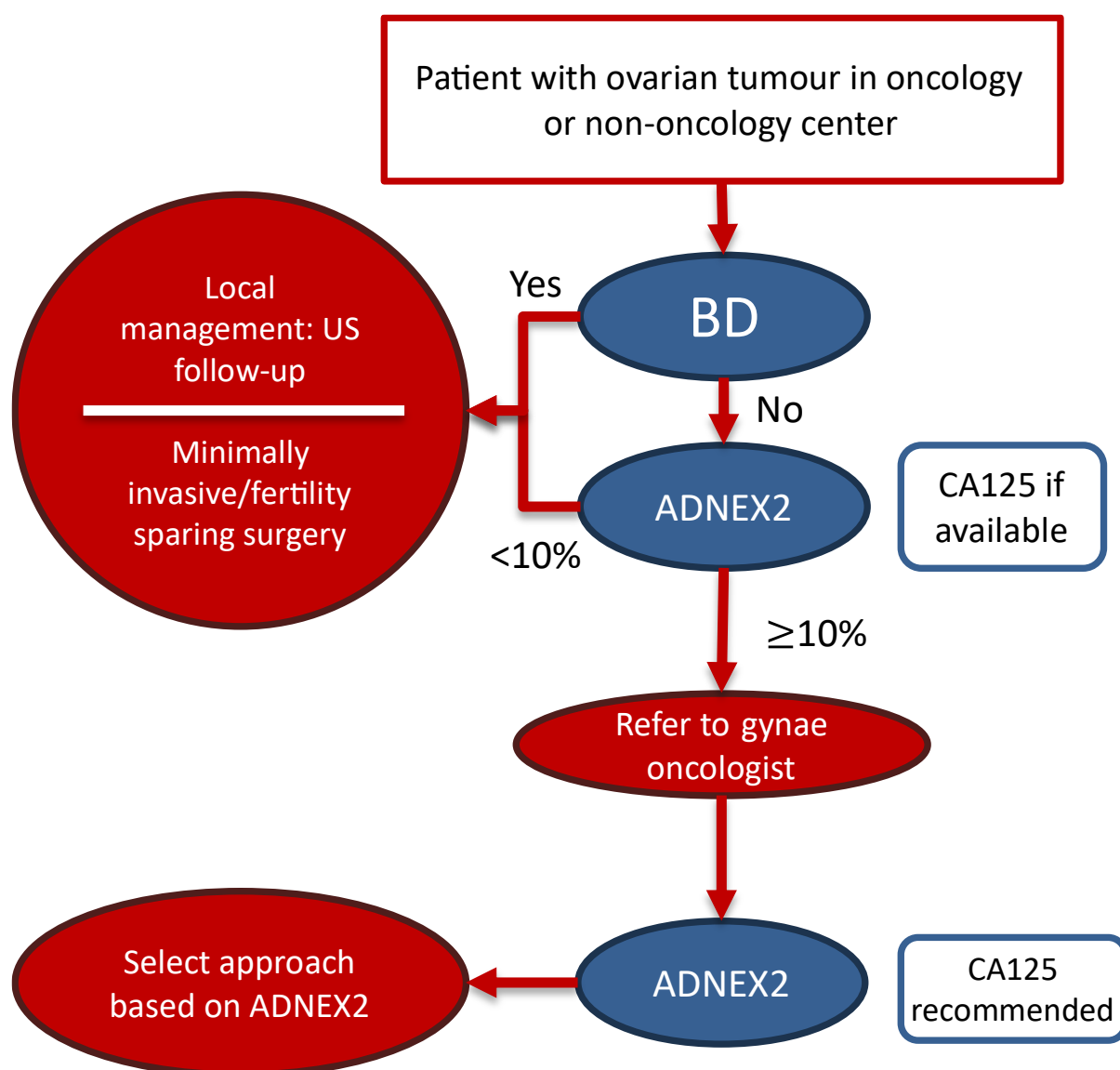


Figure 4. Patient flowchart in two-step strategy with ADNEX2

Table 1. Descriptive statistics of the study cohorts.

Variable	Benign Descriptors apply	Benign Descriptors do not apply	All patients
Number of cases	2190	4657	6847
Benign	1987 (91%)	2638 (57%)	4625 (68%)
Borderline	4 (0.2%)	387 (8%)	391 (6%)
Stage I invasive cancer	2 (0.1%)	301 (6%)	303 (4%)
Stage II-IV invasive cancer	2 (0.1%)	785 (17%)	787 (11%)
Secondary Metastasis	2 (0.1%)	227 (5%)	229 (3%)
Uncertain ^a	193 (9%)	319 (7%)	512 (7%)
Surgery without follow up scan within 120 days after first scan	726 (33%)	3174 (68%)	3900 (57%)
Patient age at recruitment (years) ^b	42 (32–54) [18–98]	54 (41–66) [18–99]	50 (37–63) [18–99]
Postmenopausal	656 (30%)	2646 (57%)	3302 (48%)
Pelvic pain	394 (18%)	594 (13%)	988 (14%)
Presence of solid components	0 (0%)	2801 (60%)	2801 (41%)
Proportion of solid components ^b	0 (0–0) [0–0]	0.25 (0–0.81) [0–1]	0 (0–0.49) [0–1]
Maximum diameter of the lesion (mm) ^b	45 (32–61) [7–99]	65 (43–105) [7–751]	56 (39–87) [7–751]
CA125 (U/ml) ^b	15 (9–30) [1–7745]	35 (14–184) [1–78200]	27 (12–127) [1–78200]
CA125 missing	1498 (68%)	1728 (37%)	3226 (47%)
Shadows ^b	378 (17%)	926 (20%)	1304 (19%)
Ascites ^b	8 (0.4%)	482 (10%)	490 (7%)
Number of papillations ^b			
None	2190 (100%)	3677 (79%)	5867 (85%)
1	0 (0%)	464 (10%)	464 (6%)
2	0 (0%)	143 (3%)	143 (2%)
3	0 (0%)	95 (2%)	95 (1%)
4 or more	0 (0%)	278 (6%)	278 (4%)
More than 10 cyst locules ^b	0 (0%)	634 (14%)	634 (9%)
Colour score of intratumoral flow			
1: No blood flow	1574 (72%)	1263 (27%)	2837 (41%)
2: minimal blood flow	449 (21%)	1314 (28%)	1763 (26%)
3: moderate blood flow	161 (7%)	1376 (30%)	1537 (22%)
4: very strong flow	6 (0.3%)	704 (15%)	710 (10%)
Tumour type using IOTA terminology			
Unilocular	2190 (100%)	388 (8%)	2578 (38%)
Unilocular solid	0 (0%)	630 (13%)	630 (9%)
Multilocular	0 (0%)	1438 (31%)	1438 (21%)
Multilocular solid	0 (0%)	1111 (24%)	1111 (16%)
Solid	0 (0%)	1060 (23%)	1060 (15%)
Not possible to classify	0 (0%)	30 (1%)	30 (<1%)

Results are presented as median (IQR) [range] for continuous variables, or N (%) for categorical variables.

^a The outcome for uncertain tumours was imputed with multiple imputation. Tumours are uncertain if there was insufficient information to make a reasonable classification of the nature of the mass at inclusion.

^b ADNEX and ADNEX2 predictors

Table 2. Summary of performance measures of ADNEX and ADNEX2 in the different populations and strategies. 10% decision threshold was used for Sensitivity, Specificity, Net Benefit and p-useful.

Model	AUC (CI)	PI	OE (CI)	Sensitivity (CI)	Specificity (CI)	Δ NB	P-useful
Where benign descriptors do not apply (N = 4657)							
ADNEX2 without CA125	0.92 (0.90 to 0.93)	(0.86 to 0.96)	1.05 (0.96-1.14)	0.95 (0.92-0.97)	0.68 (0.63-0.73)	0.0365	92%
ADNEX2 with CA125 if available	0.93 (0.91 to 0.94)	(0.88 to 0.97)	1.04 (0.93-1.14)	0.95 (0.92-0.97)	0.69 (0.64-0.74)	0.0364	91%
ADNEX without CA125	0.92 (0.90 to 0.93)	(0.87 to 0.96)	1.16 (1.08-1.24)	0.94 (0.90-0.96)	0.70 (0.64-0.75)	0.0368	89%
ADNEX with CA125 if available	0.93 (0.91 to 0.94)	(0.88 to 0.96)	1.13 (1.05-1.22)	0.94 (0.90-0.96)	0.71 (0.65-0.76)	0.0365	90%
All patients (N=6847)							
2SS+ADNEX2 without CA125	0.94 (0.93 to 0.95)	(0.89 to 0.98)	1.05 (0.95-1.14)	0.93 (0.89-0.95)	0.83 (0.78-0.87)	0.0596	98%
2SS+ADNEX2 with CA125 if available	0.95 (0.93 to 0.96)	(0.90 to 0.98)	1.04 (0.94-1.14)	0.93 (0.89-0.95)	0.84 (0.79-0.87)	0.0594	97%
2SS+ADNEX without CA125	0.94 (0.93 to 0.95)	(0.90 to 0.97)	1.16 (1.08-1.24)	0.92 (0.87-0.95)	0.84 (0.79-0.88)	0.0594	97%
2SS+ADNEX with CA125 if available	0.95 (0.93 to 0.96)	(0.91 to 0.98)	1.13 (1.05-1.22)	0.92 (0.87-0.95)	0.85 (0.80-0.89)	0.0592	98%
ADNEX2 without CA125	0.94 (0.93 to 0.95)	(0.89 to 0.98)	1.04 (0.94-1.12)	0.93 (0.89-0.95)	0.83 (0.78-0.87)	0.0596	98%
ADNEX2 with CA125 if available	0.95 (0.93 to 0.96)	(0.90 to 0.98)	1.03 (0.92-1.13)	0.92 (0.89-0.95)	0.84 (0.79-0.87)	0.0593	97%
ADNEX without CA125	0.94 (0.93 to 0.95)	(0.90 to 0.97)	1.14 (1.05-1.22)	0.92 (0.87-0.95)	0.84 (0.79-0.88)	0.0594	97%
ADNEX with CA125 if available	0.95 (0.93 to 0.96)	(0.91 to 0.98)	1.11 (1.03-1.20)	0.92 (0.87-0.95)	0.85 (0.80-0.89)	0.0593	98%

AUC: Area Under the receiver operating characteristic curve, ECI: Estimated calibration index, CI: Confidence interval, PI:

Prediction interval, OE: Observed over expected ratio, Δ NB: Delta Net Benefit calculated at the 10% decision threshold.

This is the net benefit of using the model minus the maximum net benefit of treat all or treat none, P-useful: Probability of being useful in a hypothetical new centre at the 10% decision threshold. 2SS: Two-step strategy

Table 3. Polytomous discrimination index and pairwise AUC performance of ADNEX2 and ADNEX to distinguish between outcome categories. Results are shown with 95% confidence intervals.

	Patients where BDs do not apply (N = 4657)				All patients (N = 6847)			
	ADNEX2 without CA125	ADNEX2 with CA125 if available	ADNEX without CA125	ADNEX with CA125 if available	2SS with ADNEX2 without CA125	2SS with ADNEX2 with CA125 if available	2SS with ADNEX without CA125	2SS with ADNEX with CA125 if available
PDI (95% CI)	0.66 (0.64-0.68)	0.67 (0.65-0.69)	0.66 (0.64-0.68)	0.69 (0.67-0.71)	0.74 (0.72-0.76)	0.76 (0.74-0.79)	0.74 (0.72-0.76)	0.76 (0.74-0.78)
Benign vs Borderline	0.88 (0.86-0.90)	0.88 (0.86-0.90)	0.88 (0.86-0.89)	0.88 (0.86-0.90)	0.92 (0.91-0.94)	0.93 (0.91-0.94)	0.92 (0.90-0.93)	0.92 (0.91-0.94)
Benign vs Stage I	0.93 (0.91-0.94)	0.93 (0.91-0.95)	0.93 (0.91-0.95)	0.93 (0.91-0.95)	0.96 (0.94-0.97)	0.96 (0.94-0.97)	0.96 (0.94-0.97)	0.96 (0.94-0.97)
Benign vs Stage II-IV	0.97 (0.96-0.97)	0.98 (0.97-0.98)	0.97 (0.96-0.97)	0.98 (0.97-0.98)	0.98 (0.97-0.98)	0.98 (0.98-0.99)	0.98 (0.97-0.98)	0.98 (0.98-0.99)
Benign vs Metastatic	0.95 (0.93-0.96)	0.95 (0.93-0.96)	0.94 (0.93-0.96)	0.95 (0.93-0.96)	0.96 (0.94-0.97)	0.96 (0.95-0.97)	0.96 (0.94-0.97)	0.96 (0.95-0.97)
Borderline vs Stage I	0.79 (0.75-0.82)	0.79 (0.75-0.82)	0.76 (0.73-0.80)	0.76 (0.72-0.79)	0.79 (0.75-0.82)	0.78 (0.75-0.82)	0.76 (0.73-0.80)	0.76 (0.72-0.79)
Borderline vs Stage II-IV	0.91 (0.89-0.92)	0.93 (0.91-0.94)	0.90 (0.88-0.92)	0.92 (0.90-0.94)	0.91 (0.88-0.92)	0.92 (0.91-0.94)	0.90 (0.88-0.92)	0.92 (0.90-0.93)
Borderline vs Metastatic	0.88 (0.85-0.91)	0.88 (0.85-0.91)	0.87 (0.84-0.90)	0.87 (0.84-0.90)	0.88 (0.85-0.90)	0.88 (0.85-0.90)	0.87 (0.84-0.90)	0.87 (0.84-0.90)
Stage I vs Stage II-IV	0.73 (0.70-0.77)	0.82 (0.79-0.84)	0.74 (0.70-0.77)	0.82 (0.79-0.85)	0.73 (0.70-0.77)	0.82 (0.79-0.84)	0.74 (0.70-0.77)	0.82 (0.79-0.85)
Stage I vs Metastatic	0.73 (0.69-0.77)	0.73 (0.69-0.77)	0.72 (0.67-0.76)	0.71 (0.67-0.76)	0.73 (0.69-0.77)	0.73 (0.69-0.77)	0.71 (0.67-0.76)	0.71 (0.67-0.75)
Stage II-IV vs Metastatic	0.65 (0.61-0.69)	0.80 (0.76-0.83)	0.66 (0.62-0.70)	0.80 (0.77-0.83)	0.65 (0.61-0.69)	0.80 (0.76-0.83)	0.66 (0.62-0.70)	0.80 (0.77-0.83)

2SS: Two-step strategy, BD: Benign descriptors, PDI: Polytomous discrimination index

1 INTRODUCTION

When confronted with an ovarian tumour, clinical staff has to decide on the appropriate management. The optimal management depends on the type of tumour. Most benign tumours can be followed up conservatively, whereas different types of malignant tumours require different treatment. The Assessment of Different NEoplasias in the adneXa (ADNEX) model is a mathematical model to estimate the likelihood of different tumour types: benign, borderline malignant, stage I primary invasive, stage II-IV primary invasive, and secondary metastasis (Van Calster et al., 2014). ADNEX includes three clinical and six ultrasound variables. It has one version with serum CA125 as a predictor and another one without. The other predictors are age, type of centre at which the ultrasound examination was performed (oncology centre vs other) and the following ultrasound variables: maximum diameter of the lesion, proportion of solid tissue, number of papillations, more than 10 cyst locules, ascites, and acoustic shadows. Type of centre is included to acknowledge differences in prevalence of malignant tumours between centres. ADNEX is intended to be used by clinicians after ultrasound examination of a patient with an adnexal mass to assess the risk of the mass being malignant. According to an international consensus document masses judged to be almost certainly benign (ADNEX risk lower than 1%) can be managed with clinical and ultrasound follow up. If malignancy risk is between 1 and 10% the tumour can be surgically managed in a general gynaecology unit. If risk of malignancy is 10% or higher, the patient should be referred to a gynaecologic oncology specialist, and the risk of each malignant tumour subtype can then be used to help decide the type of surgical treatment (D. Timmerman et al., 2021). Nevertheless, ADNEX can be improved for three reasons. First, the model was based on patients recruited between 1999 and 2012 (IOTA phases 1-3), i.e., 14 to 27 years ago. Populations are bound to change over time, which may negatively affect model performance (Van Calster et al., 2023). Retraining models on more recent data, if possible, is therefore of interest.

Second, like nearly all studies in the field, the ADNEX development study included only patients that underwent surgery. While this approach allows the tumour outcome to be based on a reliable reference standard (histopathology), it suffers from selection bias by not including ovarian masses that are not operated on. This is problematic for all available models: while these model are developed to support the decision to operate or not, model development samples excluded patients where the eventual decision was to not operate. Third, there is strong evidence that a large proportion of tumours can be classified as benign without a mathematical model by using the Benign Descriptors (BDs) (**Box 1**) (Landolfo et al., 2023). A two-step strategy has been proposed, where first the benign descriptors are used and then ADNEX is applied to the tumours where no BD applies (Landolfo et al., 2023). A prospective multicenter validation indicated that the BDs can be used in 37% of patients, of which 99.3% had a benign tumour. This approach has been validated seven times, showing similar performance (Barcroft et al., 2024; Borges et al., 2024; De Vitis et al., 2025; Einig et al., 2025; Pascual et al., 2024; Perry et al., 2026; S. Timmerman et al., 2023). Hence, it makes sense to update ADNEX only on data from patients for whom the BD do not apply.

Box. The benign simple descriptors.

Descriptor 1: Unilocular cyst, ground glass echogenicity, maximum lesion diameter (MLD) <10cm, premenopausal; this descriptor is suggestive of an **endometrioma**.

[Image]

Descriptor 2: Unilocular cyst, mixed echogenicity, acoustic shadows, MLD less than 10cm, premenopausal; this descriptive is suggestive of a **benign cystic teratoma**.

[Image]

Descriptor 3: Unilocular cyst, anechoic cyst fluid, smooth internal walls, MLD less than 10cm; this descriptive is suggestive of a **simple cyst or cystadenoma**.

[Image]

Descriptor 4: Remaining unilocular cysts with smooth internal walls and MLD less than 10cm.

[Image]

Assessment: If any descriptor is present, the risk of malignancy is presumed to be <1.1%.

The objective of this study is to update the ADNEX model on recent data from patients that underwent surgery or were followed conservatively and with tumours to which the BDs did not apply, and to

validate the model. Such an updated model should be better suited for use in a two-step strategy, where application of the BDs is the first step.

2 METHODS

2.1 DESIGN AND PARTICIPANTS

We use data from the IOTA 5 study, an ongoing multicentre prospective cohort study including patients with an adnexal mass managed with surgery or with clinical and ultrasound follow-up. Patients were recruited between 1st January 2012 and 31st December 2016 (D. Timmerman et al., 2025). Conservatively managed patients included here were followed up until 31st December 2022. Inclusion criteria were being at least 18 years old and presenting with at least one adnexal mass (ovarian, para-ovarian, or tubal) at recruitment. Exclusion criteria were having a lesion presumed to be physiological with a largest diameter less than 3 cm or refusing or withdrawing informed consent. Pregnancy was not an exclusion criterion. We also excluded patients with a mass that was already in follow-up at the recruitment centre before recruitment into the study.

Patients were recruited at 39 oncology and non-oncology centres across 15 countries, with oncology centres defined as centres with a dedicated gynaecologic oncology unit (**Table S1**). For this study, we use data on 6847 patients recruited at 17 centres. These 17 centres satisfied predefined quality criteria and aimed to consecutively recruit patients selected for surgery or for management with clinical and ultrasound follow-up (**Figure 1**). Criteria for excluding centres are described in **Supplementary material S1**. The IOTA5 study was approved by the Research Ethics Committee from the University Hospitals KU Leuven as the coordinating centre (B32220095331/S51375), and by local ethics committees of all

participating centres. We reported the study according to the TRIPOD+AI guideline (Collins, Moons, et al., 2024).

2.2 PROCEDURES AND VARIABLE MEASUREMENTS

Patients were recruited by ultrasound examiners. The ultrasound examiners were gynaecologists with expertise in gynaecological ultrasound. They were either IOTA trainers or had passed the IOTA certification test (<https://iotaplus.org/en/certified-members>). When recruiting, the ultrasound examiners followed a standardised research protocol (<https://clinicaltrials.gov/study/NCT01698632>). Standardized clinical information was collected and a standardised transvaginal ultrasound examination was performed supplemented with abdominal ultrasound if necessary. The recruitment examination included scanning the uterus, both adnexa, and the whole pelvis outside these organs. Grey scale and colour or power Doppler ultrasound was used to characterise the morphology and vascularisation of each adnexal mass. Information on a set of predefined ultrasound variables, including those used in the ADNEX model, was collected, and the IOTA terminology was used to describe the ultrasound findings (D. Timmerman et al., 2000). In addition, the ultrasound examiners provided a diagnosis based on their subjective assessment by classifying each lesion as benign, borderline, or malignant, and by specifying the degree of certainty of their classification (certain, probable, or uncertain). Based on pattern recognition they also suggested a specific histology for each mass choosing from a list of 18 predefined diagnoses (Valentin, 2004).

When multiple adnexal masses were found on ultrasound, the dominant mass was included in the statistical analysis. The dominant mass was the mass with the most complex ultrasound morphology. If multiple masses had similar morphology, the largest mass or the mass that was most easily accessible with ultrasound was denoted dominant. We encouraged centres to measure the patients' serum CA125 level, but this was not a requirement for inclusion in the study. Measurement of CA125 was left

to clinical judgement and local protocols. Due to the prospective study design, predictor measurements were blinded to tumour outcome.

Based on patient history, clinical and ultrasound findings, the ultrasound examiner suggested surgery or conservative management with follow-up. The final decision on management was made by the treating physician in dialogue with the patient. Conservative management included ultrasound and clinical follow-up at intervals of three months, six months, and then every 12 months thereafter. At follow-up visits clinical information, including symptoms, was collected, an ultrasound examination was performed and the same standardized ultrasound information as at the inclusion scan was collected. After one or more follow-up visits, some patients underwent surgery for a variety of reasons (e.g., suspicion of malignancy or patient anxiety) (Froyman et al., 2019). For some patients, the mass resolved spontaneously during follow-up. Each centre performed surgery by following local protocols and histological examination of surgically removed masses.

2.3 OUTCOME

Patients who were managed conservatively were followed up until the first occurrence of one of the following events: the mass resolved spontaneously, the mass was surgically removed, the patient died, or the date was 31 December 2022. The outcome describes the nature of the adnexal mass at inclusion (benign, borderline, stage I primary invasive, Stage II-IV primary invasive, or secondary metastasis). The primary outcome is classification of tumours as benign or malignant at inclusion. This classification is based on histology when patients underwent surgery, or on subjective assessment of ultrasound images at inclusion and during follow-up until 12 (+/-2) months if surgery was not performed (Van Calster et al., 2020). If there was insufficient information to make a reasonable classification of the nature of the mass at inclusion the outcome was classified as uncertain. **Table S4** shows the classification criteria.

Pathologists were blinded to ultrasound predictor variables and model predictions but might have received information on subjective assessment by the ultrasound examiner when clinically relevant. Borderline tumours were classified as malignant. Central pathology review was not done because in a previous study we did not observe important differences in reported outcomes between local and central pathology reports (D. Timmerman et al., 2005). The stage of the malignant tumours was determined using the criteria by the International Federation of Gynaecology and Obstetrics (FIGO) (A.P.M. Heintz et al., 2003).

2.4 DATA CLEANING AND PREPARATION

Patient data was collected using a study-specific secure electronic platform (Astraia software, Munich, Germany), and encrypted data communication to ensure data security. As of x/x/202x, data were collected through a different but similar platform, Clinical Data Miner (CDM) (Installé et al., 2014). A team of biostatisticians and ultrasound examiners performed data cleaning through regular meetings and by sending queries to participating centres to retrieve missing information or to correct inconsistencies.

2.5 STATISTICAL ANALYSIS

We compared the available sample size with Pate et al's sample size criteria for multinomial logistic regression prediction models (Pate et al., 2023), with details provided in **supplementary material S2.1**. We followed a prespecified statistical analysis plan for this study preregistered in Open Science Framework (https://osf.io/36n8m/overview?view_only=0f1fa6d9a5e34e4d901c05922d0d0307).

Missing values occurred for CA125 and the outcome (when labelled as uncertain), and were handled with multiple imputation (**Table S4**). Details about missing data handling are found in **supplementary material S2.2**.

ADNEX without CA125 and ADNEX with CA125 were updated on the dataset of patients with tumours to which the BDs did not apply. We used the following procedure to obtain ADNEX2. We used the same predictors and transformations (for continuous predictors) as in ADNEX. First, **we fitted a multinomial logistic recalibration model to update intercepts and coefficients using slopes** (Van Calster et al., 2017). In the second step, **we used Bayesian refitting using information from the original ADNEX model and the logistic recalibration** (Su et al., 2018; Van De Schoot et al., 2021). **In both steps, we used models with random intercepts per centre** (Wynants, Vergouwe, et al., 2018). Details on the model updating procedure are found in **supplementary material S2.3.1**.

To validate the two-step strategy, patients with tumours to which the BDs applied were assigned risk estimates for the five outcome categories. These estimates were obtained as the prevalence of each outcome category in the subgroup of patients with tumours to which the BDs applied, averaged over the 100 imputed datasets.

We validated the two-step strategy and ADNEX2 (with and without CA125) using internal-external cross-validation (IECV) to estimate how the model would perform in new centres (Collins, Dhiman, et al., 2024, p. 1; Royston et al., 2004). Internal-external cross-validation mimics geographical external validation by treating every centre once as a test set for models developed on the remaining 16 centres. The approaches to update ADNEX2 and to generate risk estimates when the BDs apply were implemented on data from 16 centres and tested on data from the remaining centres. To validate ADNEX2 with CA125, in isolation or as part of the two-step strategy, we used a realistic approach by using ADNEX2 with CA125 when CA125 was available and ADNEX without CA125 when CA125 was not available.

We present centre-specific results of the IECV procedure, as well as overall results using meta-analysis of centre-specific results, similar to previous work (Eriksson et al., 2020; Van Calster et al., 2014). 95% confidence intervals (CI) are reported. Heterogeneity in performance between centres is quantified with prediction intervals (PI) (Partlett & Riley, 2017). We report diagnostic performance for distinguishing benign from malignant tumours as follows: distribution of estimated risk per outcome category, area under the Receiver Operating Characteristic (ROC) curve (AUC), flexible centre-specific and overall calibration plots for the overall estimated risk of malignancy, and net benefit with decision curves for a decision threshold of 1% (to decide whether conservative management should be considered) and for decision thresholds between 5 and 40% (to decide whether the patient requires specialised care) on the estimated risk of malignancy (Barreñada, Campo, et al., 2025; Debray et al., 2019; Van Calster et al., 2025; Wynants, Riley, et al., 2018). Sensitivity, specificity, positive predictive value and negative predictive value for the same decision thresholds were also calculated. We report diagnostic performance for discriminating between the five subtypes of adnexal masses as polytomous discrimination index (PDI), pairwise AUCs and flexible calibration plots, all while ignoring the multicentre nature of the data due to low numbers of centre-specific events for some outcome categories (Van Calster, Van Belle, et al., 2012; Van Calster, Vergouwe, et al., 2012; Van Hoorde et al., 2014). Methodological details about the performance measures are available in **Supplementary material S2.3**.

We analysed performance for pre-specified subgroups: (1) postmenopausal versus premenopausal patients, (2) oncology centres versus other centres, and (3) patients that were operated within 120 days after inclusion scan. In the subgroups, only results for the two-step strategy are shown.

2.6 INDEPENDENT EXTERNAL VALIDATION

The recalibrated and refitted coefficients of ADNEX2 are not shown because they will be included in a CE-labelled app offered by Gynaia Inc. (<https://www.gynaia.com/>). Members of the ovarian programme of Gynaia will have access to the app to use it as a decision support tool.

Researchers that aim to externally validate the model can do it by using *evaluatr* R package [link]. This package provides a safe environment for model developers and model evaluators where the developer does not have access to the patient data, and the evaluator does not have access to the model coefficients but can obtain a vector of predicted risks to evaluate the model's aggregate performance. To use *evaluatr* a minimum number of X events is needed. Access to the model is given case by case, upon reasonable request to the corresponding author, including the purpose and protocol for the model validation.

3 RESULTS

After exclusions and data cleaning, 6847 patients recruited at 17 centres were included in the analysis (**Figure 1, Table S1, Supplementary material S1**). The median age was 50 years (interquartile range 37-63) and 3302 patients (48%) were postmenopausal (**Table 1**). The tumour was benign in 4625 (68%) patients, malignant in 1710 patients (25%), and uncertain in 512 (7%) patients. Of the malignant tumours, 391 (6%) were borderline, 303 (4%) stage I primary invasive, 787 (12%) stage II-IV primary invasive, and 229 (3%) secondary metastatic. The BDs applied to 2190/6847 (32%) tumours. The sample of 4657 patients with tumours to which a BD did not apply was the development cohort for ADNEX2. Here, median age was 54 years (interquartile range 41–66), and 2646 patients (57%) were postmenopausal (**Table 1**). 2638 patients (57%) had a benign tumour, 1700 (37%) a malignant tumour, and 319 (7%) a tumour of uncertain outcome. The serum CA125 levels was missing in 1728 patients (37%). 3174 patients (68%) underwent surgery without a follow-up scan. Further information on tumour outcome, missing CA125 values, and management decisions is provided in **Tables S5-S10**.

Table S11 provides descriptive statistics of the 4657 patients on which ADNEX2 was developed and the 5909 patients on which ADNEX was developed. The main difference between both cohorts was the higher proportion of unilocular tumours in ADNEX (30%) with respect to ADNEX2 (8%) and the higher presence of shadows with 13% in ADNEX and 20% in ADNEX2 (**Table S11**).

3.1 BENIGN DESCRIPTORS

2180 of the 2190 tumors for which a BD applied (99.5%) were benign or had an uncertain outcome, 10 (0.5%) were malignant (4 borderline, 2 stage I invasive, 2 stage II-IV invasive, 2 secondary metastatic) (**Table S5**). After multiple imputation of uncertain outcomes, 99.45% tumours (99.1% - 99.8%) were benign, 0.21% (0.01% - 0.42%) borderline, 0.10% (<0.01% - 0.25%) stage I primary invasive, 0.12% (<0.01% - 0.28%) stage II-IV primary invasive, and 0.11% (<0.01% - 0.26%) secondary metastatic (**Table S6**).

3.2 ADNEX2

Figures S4-S6 present histograms of the predictions of the 6 validations per binary outcome and **Figures S7-S9** the risk distribution per tumour subtype. The median highest estimated risk was always the true outcome, except for metastatic cancers in the model without CA125 where Stage II-IV and metastatic had similar estimated risks. The relation between predicted risks and variables was very similar for ADNEX and ADNEX2 (**Figures S10-S11**) but ADNEX2 tended to predict higher risks of malignant subtypes, especially borderline and metastatic cases (**Figures S12-S13**).

3.2.1 Discrimination

The AUC to discriminate between benign and malignant tumours in all patients was 0.94 (95% CI 0.93-0.95, 95% PI 0.89-0.98) for the two-step strategy using ADNEX2 without CA125 and 0.95 (95% CI 0.93-0.96, 95% PI 0.90-0.98) for the two-step strategy using ADNEX2 with CA125 if available (**Table 2, Table S12, and Figure S14**). On the patients for which the BDs did not apply, the AUC was 0.92 (0.90-0.93) for ADNEX2 without CA125 and 0.93 (0.91-0.94) for ADNEX2 with CA125 if available. The AUCs for ADNEX and ADNEX2 whether used alone or in a two-step strategy were similar. Using ADNEX2 with CA125 if available slightly improved AUC over the version without CA125. Forest plots with centre-specific performance are shown in **Figures S15-S26**.

ADNEX2 with CA125 if available performed better than ADNEX2 without CA125 to discriminate stage II-IV primary invasive from other malignant subtypes (**Table 3**). Importantly, ADNEX2 discriminated better than ADNEX between borderline and stage I tumours.

3.2.2 Calibration

ADNEX2 was well calibrated and showed better calibration than ADNEX for predicting risk of malignancy whether used alone or in a two-step strategy. However with slightly wider prediction intervals (**Figure 3, Figures S27-S38**). O:E ratios were between 1.03 and 1.05 for ADNEX2 and between 1.11 and 1.16 for ADNEX (**Table 2 and Table S13**). This reflects the better calibration for ADNEX2: ADNEX2 underestimated risks less than ADNEX (3 to 5% versus 11 to 16%). Calibration plots for each of the tumor subtypes confirmed that ADNEX2 was better calibrated than ADNEX (**Figures S39-S42**).

3.2.3 Classification and clinical utility

When using the two-step strategy with ADNEX2, the 10% risk of malignancy threshold yielded sensitivity 93% and specificity 83-84%, irrespective of whether ADNEX2 without CA125 or ADNEX2 with CA125 if available was used (**Table 2**). Using the 1% threshold, sensitivity was 97% and specificity 52% irrespective of whether ADNEX2 was used with or without CA125 (**Table S14**). The classification performance when using ADNEX was similar with sometimes slightly lower sensitivity and slightly higher specificity.

The two-step strategy with ADNEX2 and that with ADNEX had similar net benefit for decision thresholds up to around 10% (**Figure 4, Figures S43 and S44**). At 1% decision threshold the models showed benefit but no or minimal utility with respect to the decision of treat all patients. The models were clinically useful for the rest of decision thresholds. At decision thresholds from 15% to 35%, using ADNEX2 instead of ADNEX in the two-step strategy had higher net benefit. At higher thresholds ADNEX2 had higher net benefit only when CA125 was not used (**Table 2, Table S15, Figures S43 and S44**). The meta-analysis estimated that, at the 10% threshold, the two-step strategy with ADNEX2 was useful in 97% of centres (CA125 used if available) or 98% of centres (without CA125), whereas the two-

step strategy with ADNEX was useful in 98% of centres (CA125 used if available) and 97% of centres (without CA125). Results for all decision thresholds are available in **Table S15**, decision curves, relative utility curves and curves for the probability of being useful are shown in **Figures S43-S48**.

3.2.4 Subgroup analysis

Predefined subgroups were based on menopausal status (premenopausal vs postmenopausal), type of centre (oncology vs non-oncology) and a subgroup defined as patients that were operated within 120 after inclusion scan. The AUCs of the two step strategy with ADNEX2 or ADNEX ranged from 0.91 to 0.95 (**Figure S49**). AUCs were higher in postmenopausal than in premenopausal patients and in oncology than in non-oncology centres. Pairwise AUCs in the subgroup of patients operated within 120 days showed similar performance to the performance in the whole population (**Table S16**). The discrimination of the two step strategy with ADNEX2 or ADNEX showed identical results. Forest plot for the two step strategy with ADNEX2 in all subgroups are included in **Figures S50-S59**. Predicted risks were underestimated across all subgroups, with slightly better calibration in non-oncology centres (**Figures S60-S79**). Calibration was more heterogeneous between centres among oncology centres than among non-oncology centres, as indicated by the wider prediction intervals (**Figures S72-S79**). Calibration was better but slightly more heterogeneous for ADNEX2 than for ADNEX across all subgroups. The number of cases in the subgroups was too low to perform multinomial calibration.

For the 10% threshold, sensitivity ranged between 0.85 and 0.96 and specificity between 0.72 and 0.89 (**Table S17**). For the 1% threshold, the sensitivity ranged between 0.91 and 0.99 and specificity between 0.31 and 0.61 (**Table S17**). The two-step strategy with ADNEX2 showed clinical utility across the whole range of decision thresholds from >1% to 40% in all subgroups, with comparable performance irrespective if CA125 was used or not (**Table S18, Figure S80**). However, at the 1% decision threshold the utility was lower than the strategy of treating all patients. The proportion of centres in which the two-step strategy has clinical utility varied between 45% and 84% for the 1% decision threshold and between 89% and 97% for the 10% decision threshold. Clinical utility was higher

in postmenopausal patients than premenopausal patients and in oncology centres rather than non-oncology centres. Clinical utility of the two-step strategy with ADNEX was similar to with ADNEX2 across all subgroups.

4 DISCUSSION

4.1 PRINCIPAL FINDINGS

This study presents the development and validation of ADNEX2, an updated version of the ADNEX model. ADNEX2 presents three main improvements over the original ADNEX. First, it is developed using recent data, thereby addressing potential population drift (Van Calster et al., 2023). Second, the cohort includes patients managed conservatively via ultrasound follow-up and excludes patients where the mass is clearly benign as indicated by the benign descriptors. The inclusion of conservatively management tumours is critical, as a key purpose of the model is to identify candidates for follow-up (risk <1%). Conversely, the exclusion of clearly benign cases improves clinical relevance, as models are rarely required when the diagnosis is subjectively obvious where the benign descriptors can correctly classify almost all masses (Landolfo et al., 2023). Third, updating ADNEX on a sample that included both operated and non-operated patients, such as in the IOTA5 study (D. Timmerman et al., 2025), removes the selection bias of only including surgically managed patients and its potentially negative consequences.

The performance of the two-step strategy with ADNEX2 has shown excellent results. The model achieves AUCs of 0.94-0.95 depending on if CA125 is used or not (**Table 2**). CA125 mainly improved the ability of the model to differentiate between Stage I and Stage II-IV cancers and between Stage II-IV and second metastatic cancers (**Table 3**). [Here probably we can add some clinical context on this]. With respect to calibration, the model showed slight underestimation of risks (**Figure 2**). However,

centre specific performance was heterogeneous. This is a common issue in calibration (Amsterdam, 2024; Gupta et al., 2021; Steyerberg et al., 2019; Van Calster et al., 2020), and may be caused for a wide range of reasons: different case-mix, different clinical guidelines etc.

4.2 STRENGTHS AND LIMITATIONS

The main strength of this study is the alignment of an already successful prediction model with how it is used in clinical practice. ADNEX2 is the first model to assess ovarian masses preoperatively including patients that were managed with ultrasound follow-up. This approach reduced selection bias based on having histopathology. Other strengths are the large sample size and the multicenter nature of the study. We validated the model using IECV (also referred to as leave centre out cross-validation) and then retrained in the whole population. Therefore, we presented externally validated model performance while still using all available information for ADNEX2 development (de Hond et al., 2023).

A limitation of including patients managed through ultrasound follow-up is that it induces differential verification bias because histopathology is not available for everyone. We determine the outcome of tumours without histopathology based on results of ultrasound follow-up and on multiple imputation (Van Calster et al., 2020). We argue that differential verification bias is much less problematic than selection bias by excluding patients managed with ultrasound follow-up.

A second limitation is the presence of missing values for serum CA125. Measuring CA125 was encouraged, but not mandatory and left to local judgement. When developing ADNEX2, we handled the missing values with multiple imputation to develop ADNEX2 both with and without CA125 as a predictor. However, as opposed to other external validations, we validated ADNEX2 with a 'CA125 if available' strategy. This strategy better reflects model use in clinic, where CA125 is not available for all patients.

A third limitation is that the performance of ADNEX is slightly more optimistic than the performance of ADNEX2. We compared the IECV results of ADNEX2 with ADNEX, indicating that ADNEX2 has slightly

better calibration. For ADNEX2, every centre-specific result is a geographically validated result, whereas 11/17 centres were part of the ADNEX development such that for 11 centres the ADNEX result is a temporally validated result.

A last limitation is that model fairness could only be evaluated in terms of menopausal status, because IOTA5 does not include information about ethnicity or socioeconomic status.

4.3 COMPARE WITH OTHER STUDIES

The reported AUC of two-step strategy with ADNEX2 of 0.95 compares favourably with recent studies. A recent systematic review with meta-analysis of ADNEX external validation studies found that only two studies validated ADNEX on cohorts of patients that were managed either with surgery or with ultrasound follow-up (Barreñada et al., 2024). Both for ADNEX with and without CA125, a summary AUC for benign versus malignant tumours of 0.94 was reported based on 4,509 tumours from 17 centres for ADNEX without CA125 and on 5,132 tumours from 18 centres for ADNEX with CA125 (Barreñada et al., 2024). 95% PI was 0.82-0.98 for ADNEX without CA125 and 0.88-0.99 for ADNEX with CA125. Later, the ROcKeTS study validated ADNEX in a cohort of patients from primary or secondary care with non-specific systems and either elevated CA125 or abnormal ultrasound that were either managed surgically or by ultrasound follow-up. ROcKeTS reported an AUC of 0.83 (95% CI 0.78-0.87) in premenopausal patients and of 0.89 (0.86-0.91) in postmenopausal patients (Sundar et al., 2024, 2026). In recent head-to-head comparisons, ADNEX outperformed Risk of Ovarian Malignancy Algorithm (ROMA) and RMI in premenopausal and postmenopausal patients (Barreñada, Ledger, et al., 2025; Sundar et al., 2024, 2026). The performance of the two-step strategy has been validated seven times (Barcroft et al., 2024; Borges et al., 2024; De Vitis et al., 2025; Einig et al., 2025; Pascual et al., 2024; Perry et al., 2026; S. Timmerman et al., 2023), with studies reporting AUCs of 0.91-0.95, in line with our results.

Deep learning on ultrasound images has emerged in the last years. In a systematic review, forty studies developing image-based models to estimate the risk of malignancy of ovarian tumours were found

(Geysels et al., 2025). AUCs between 0.75 and 0.99 were reported at internal validation, however 95% studies were at high risk of bias. Most of these models were built only on patients that underwent surgery. One of these models has been externally validated in a prospective multicenter cohort, showing promising results (Christiansen et al., 2025). The study reported an AUC of 0.93 with very good calibration, but no measures of clinical utility were presented. (Van Calster et al., 2025)

4.4 RESULTS IN CONTEXT (USABILITY)

ADNEX2 is intended to be used as part of a two-step strategy (**Figure 4**). When a patient presents with a newly detected mass, the BD are first checked. If a descriptor applies, the estimated risk of malignancy is <1% and ADNEX2 is not needed. The patient can be considered for conservative management. If the BD do not apply, ADNEX2 is applied. At this point, we argue that CA125 measurement is not mandatory and the ADNEX2 version without CA125 can be used. If CA125 is available, however, we do recommend using the version with CA125. If the model's overall estimated risk of malignancy is <1%, the patient can be considered for conservative management. If the estimated risk is at least 1% but less than 10%, management using standard surgical techniques can be considered locally. If the estimated risk is at least 10%, we recommend that the patient is referred to an oncology centre for advanced gynaecological care. There, the estimated risks of the four subtypes of malignancy should be considered to guide specific management. In this case, we recommend measuring CA125 if it was not yet available, and to obtain updated risks from the ADNEX2 version with CA125 as a predictor. Depending on the local context or preferences, the threshold of 10% can be changed. If a patient is first seen in a non-oncology centre and then referred to an oncology centre, the type of centre predictor should change accordingly.

To summarise, the two-step strategy involving BD and ADNEX2 supports two decisions: the decision whether or not the patient should be managed conservatively and the decision whether or not the patient should be referred to a specialised centre for management by a gynaecological oncologist (**Figure 4**).

The model can be externally validated freely using the *evaluatr*, which is a platform developed to ensure statistical disclosure control and provide aggregate performance without compromising the model coefficients or the patient's characteristics. However, to be safely used in a decision support tool in clinic, the model needs to be used through the application developed by Gynaia Inc, complying with local regulations.

4.5 FUTURE RESEARCH

More prospectively collected multicentre studies are needed to validate the performance of ADNEX2 in patients that are surgically managed or managed with ultrasound follow up. Validating ADNEX2 in cohorts where patients are surgically managed could also be useful, but we anticipate the model to underestimate risks in that subpopulation and the performance not reflecting the performance as used in clinic.

4.6 CONCLUSION

ADNEX2 is an important improvement over ADNEX in the model-aided management of adnexal masses. The newly presented model is developed to work within the two-step strategy and including patients that are conservatively managed, improving calibration and reducing selection bias.

5 OPEN SCIENCE

Funding

[Please add your funding]

Conflicts of interest

[Please add your COI]

D.T. and T.B. are co-founders of and have stock options in Gynaia Inc., a company providing teaching, training, data collection, and decision support tools (such as the ADNEX2 model) for clinicians and researchers in the field of gynecology. B.V.C. and W.F. have stock options in Gynaia Inc. B.V.C. and D.T. report consultancy work done by KU Leuven to help implement and test the original ADNEX model in ultrasound machines by Samsung Medison, Mindray, ESAOTE, and GE Healthcare, outside the submitted work. T.B. reports grants, personal fees, and travel support from Samsung Medison; travel support from Roche Diagnostics; and personal fees from GE Healthcare, all outside the submitted work.

Protocol

The IOTA5 protocol was registered in clinicaltrials.gov (<https://clinicaltrials.gov/study/NCT01698632>, supplementary file X) and the statistical analysis plan was preregistered in an Open Science Foundation (OSF) preregistration.

Data Sharing

Follow-up data collection for the IOTA5 study is ongoing. When the study has closed and results have been published, the study responsible may share data upon reasonable request to dirk.timmerman@uzleuven.be. The R code and a dataset that includes centre code, outcome and patient risk of malignancy for each of the different models and strategies is shared to allow readers to reproduce the results and figures.

Code sharing

All code to produce the results and visualisations in the document is shared in an OSF repository [link].

Patient and public involvement

Refer to Sonia Rademakers from Esperanza.

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